Measuring Shipping Disruptions with Satellite Data

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By Ricardo Marto

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KEY TAKEAWAYS

- Satellite vessel-tracking data allow us to measure how disruptions such as port congestion, canal bottlenecks, rerouting and slower sailing reduce the effective capacity of U.S.-bound container shipping.
- During the 2021-22 supply chain crisis, carriers added capacity to U.S.-bound routes, but shipping utilization fell from about 89% before the pandemic to below 70%.
- The decline was particularly large on the Asia-to-U.S. West Coast route, with shipping utilization starting to decline in November 2020, before port congestion was visible, from an average of 92% before the pandemic to 53% by late 2021.
- In 2023, rerouting around the Cape of Good Hope following Houthi attacks in the Red Sea sharply reduced shipping utilization on routes from Asia and the Middle East to the U.S. East Coast, where utilization fell by about 21 percentage points in a year in the route going through the Suez Canal from Asia.

Recent years have seen a succession of disruptions to global shipping. During the COVID-19 pandemic, ships lined up outside the ports of Los Angeles and Long Beach, waiting to anchor to unload their cargoes; for example, on Oct. 21, 2021, 169 vessels were in the ports' area, with 109 of them waiting at anchor.[1] In March 2021, the container ship Ever Given blocked the Suez Canal for six days. A drought restricted traffic through the Panama Canal in 2023 and 2024. And attacks on commercial vessels in the Red Sea have led many ships to avoid the Suez Canal and sail around the Cape of Good Hope instead.

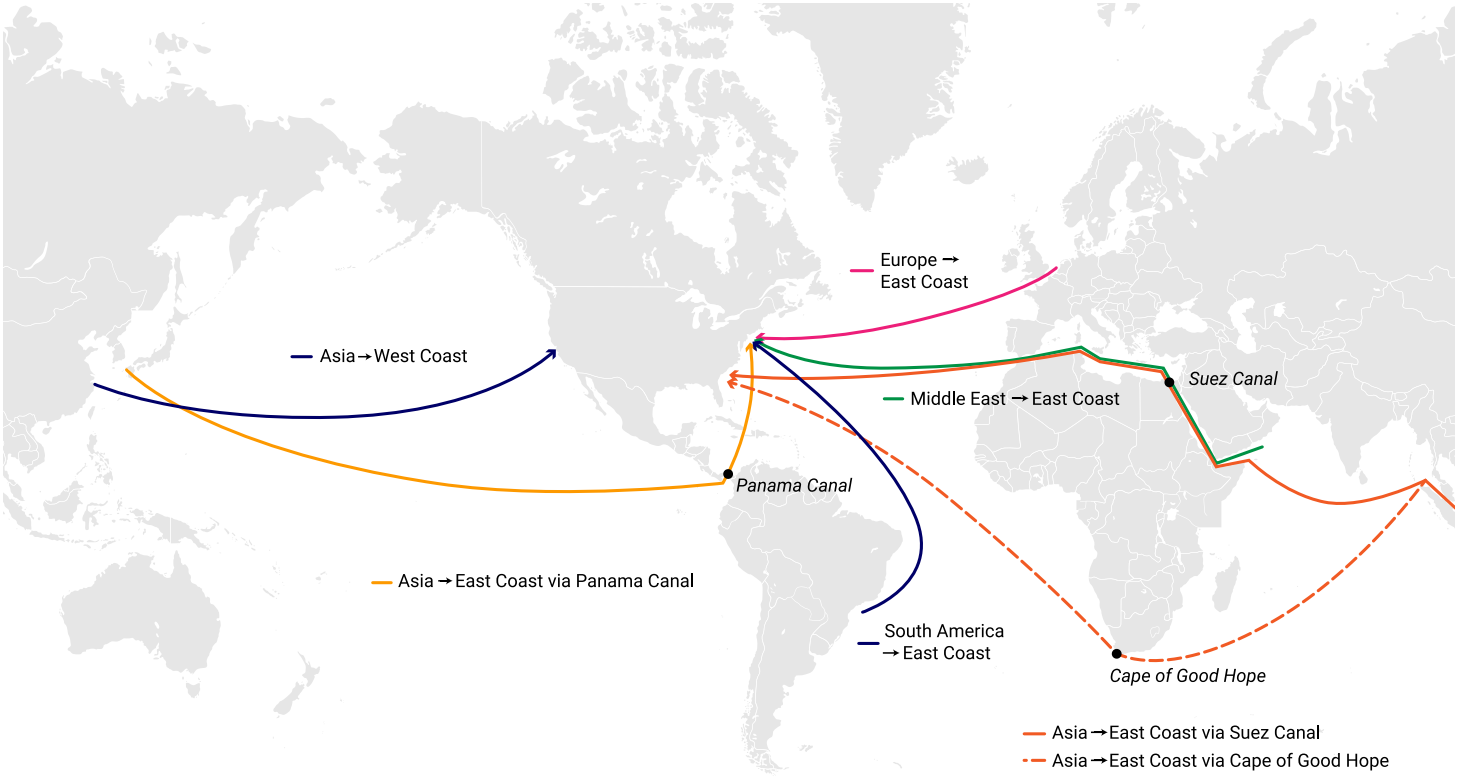
How can we measure the severity of these very different disruptions? Measures such as the number of ships waiting outside a port or the number of days required to cross an ocean can capture particular disruptions. But they are difficult to compare across routes and over time. This blog post uses satellite data covering U.S.-bound container ships to construct a common measure of how much shipping capacity is lost to different types of disruptions. The post draws on a recent working paper I wrote with Xiwen Bai, Jesús Fernández-Villaverde, Yiliang Li and Francesco Zanetti.[2] (For a summary of the working paper, view this brief video created by my co-authors and me.)

Tracking Container Ships from Space

In our working paper, my co-authors and I used Automatic Identification System, or AIS, data from 2016 through early 2025. Ships regularly transmit AIS signals containing information about their location, speed and identity. We combined these signals with vessel characteristics to reconstruct nearly 59,000 container-ship trips to the United States.

We divided these trips into six major routes connecting Asia, Europe, the Middle East and South America to the U.S. East and West coasts. Shipments from Asia to the East Coast are split into two routes: one through the Panama Canal and one through the Suez Canal. Together, these routes account for about 80% to 90% of U.S. containerized maritime imports. (The map below also displays the alternative route via the Cape of Good Hope, which was used when ships could not be routed through the Suez Canal following the Houthi attacks.)

Six Major Routes Carry Most U.S. Container Imports



NOTES: Together, these six routes (solid lines) account for more than 80% of U.S. containerized maritime imports; a seventh route (dashed line) represents the alternative route around Cape of Good Hope. Routes are stylized.

SOURCE: Bai, Fernández-Villaverde, Li, Marto and Zanetti (2026).

For each route, we first measured potential shipping capacity: the amount of shipping service that the vessels deployed on that route could provide under normal operating conditions. But having ships available does not necessarily mean that all their capacity is being used to transport goods. Ships can spend days waiting outside congested ports or canals. They can be forced to take longer routes. They can also travel more slowly. We therefore constructed a second measure, effective shipping capacity in ton-miles, which subtracts the capacity lost to four types of disruptions: port congestion, canal congestion, rerouting and slower sailing.

The ratio of effective-to-potential capacity gives us a measure of shipping utilization. A utilization rate of 90%, for example, means that disruptions are absorbing 10% of the potential shipping services that the deployed fleet could otherwise provide. This measure allows us to express very different disruptions in the same units and compare them across routes and over time.

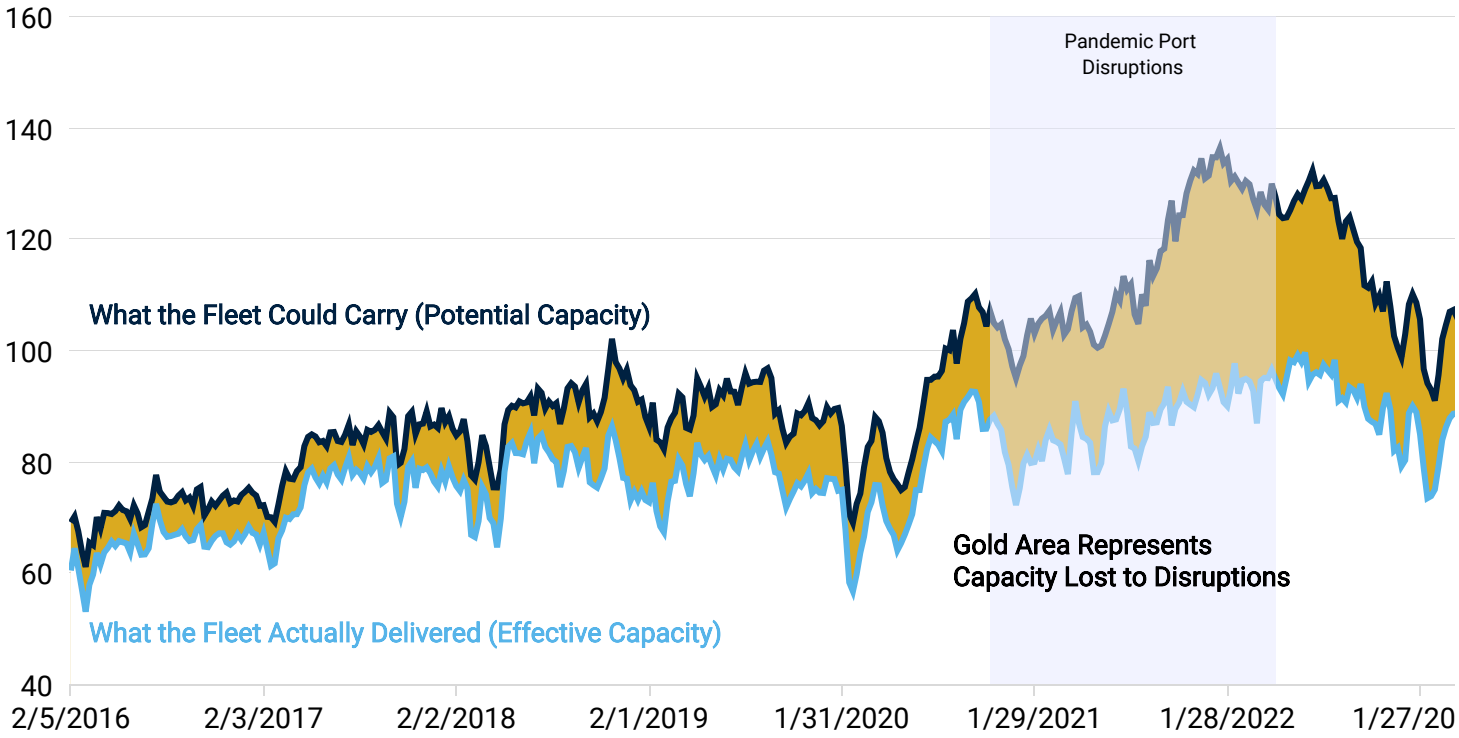
The 2021 West Coast Port Crisis

The COVID-19 pandemic supply chain crisis illustrates why the distinction between potential and effective capacity matters. As U.S. demand for imported goods surged during the pandemic, shipping companies responded by sending more ships to the United States. By early 2022, the combined capacity of the ships deployed on U.S.-bound routes had increased by more than 60% relative to its prepandemic level.

But much of this additional capacity did not translate into additional shipping services. Before the pandemic, the U.S.-bound container fleet converted about 89% of its potential capacity into effective capacity. During the 2021-22 supply chain crisis, utilization fell below 70%. At the worst point of the disruption, nearly one-third of the potential capacity deployed toward the United States was being absorbed by operational disruptions.

More Ships Do Not Mean More Shipping

Shipping Capacity on U.S.-Bound Ships, Billions of Ton-Miles per Week



SOURCE: Bai, Fernández-Villaverde, Li, Marto and Zanetti (2026).
NOTES: Weekly data. A ton-mile is one metric ton of cargo carried one nautical mile.

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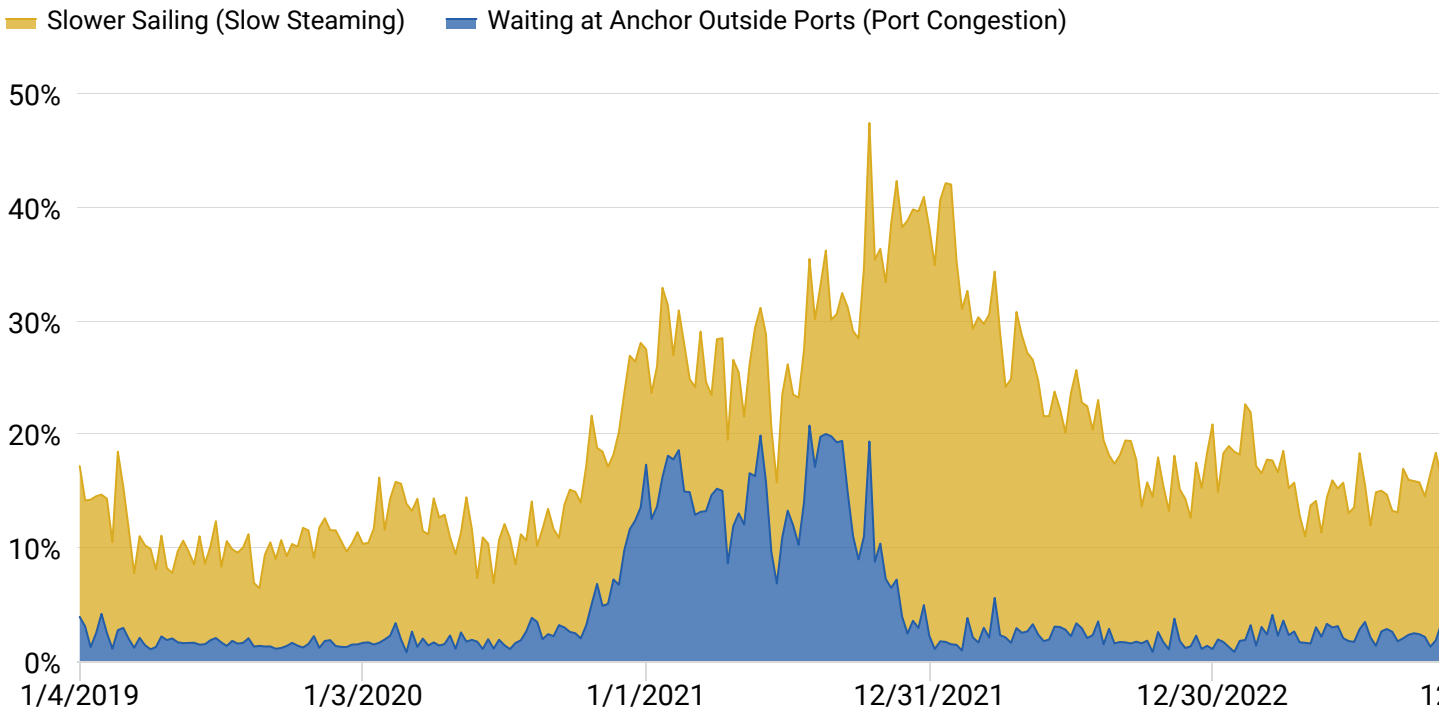
The decline was particularly severe on the route connecting Asia to the U.S. West Coast. Before 2020, this was the most highly utilized route in our data, converting about 92% of potential capacity into effective shipping services. By late 2021, utilization had fallen to 53%.

Port congestion was an important part of the story. Before the pandemic, container ships in our sample spent about six hours at anchor on average. During the summer and fall of 2021, average waiting times on the Asia-West Coast route repeatedly exceeded 60 hours. At its peak, port congestion alone absorbed about one-fifth of the route's potential capacity. But congestion at the ports was not the only source of lost capacity. Ships also slowed down substantially. In late 2021, the ports of Los Angeles and Long Beach introduced a virtual queuing system that assigned queue positions by departure time, removing the incentive to race across the Pacific only to wait at anchor. Average speeds on the route fell from about 16 knots before the pandemic to about 10 knots around the beginning of 2022. On the Asia-West Coast route, slower sailing absorbed nearly 40% of potential capacity at the height of the supply chain crisis. Together, port waiting and slower sailing absorbed nearly half of the route's potential capacity.

The satellite data also show that the deterioration began before the images of ships waiting outside Southern California ports became a familiar sight. Shipping utilization on the Asia-West Coast route began falling by November 2020, months before port congestion became especially visible in 2021.

Waiting at Port, Then Sailing Slowly

Share of Lost Potential Capacity on Asia to U.S. West Coast Route



SOURCE: Bai, Fernández-Villaverde, Li, Marto and Zanetti (2026).
NOTES: Weekly data. Before the pandemic, ships spent about six hours at anchor; in 2021, waits often exceeded 60 hours. In late 2021, a virtual Long Beach let ships slow down rather than wait at anchor.

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The Red Sea Crisis

The Red Sea crisis provides a very different example of a shipping disruption. Following the Houthi attacks on commercial vessels beginning in late 2023, many container ships stopped using the Suez Canal and instead rerouted around the Cape of Good Hope. On Dec. 30 and 31, 2023, for example, the container ship Maersk Hangzhou was attacked twice: first by a missile and then by Houthi fighters in small boats who tried to board it. Days later, Maersk announced that its ships would avoid the Red Sea for the foreseeable future. Unlike in the 2021 port crisis, the main problem was not ships waiting outside U.S. ports; it was that ships had to travel substantially farther to deliver the same goods.

Our satellite data allow us to observe these changes in vessel trajectories directly. Rerouting around the Cape of Good Hope meant that ships sailing from Asia to the U.S. East Coast covered roughly 600

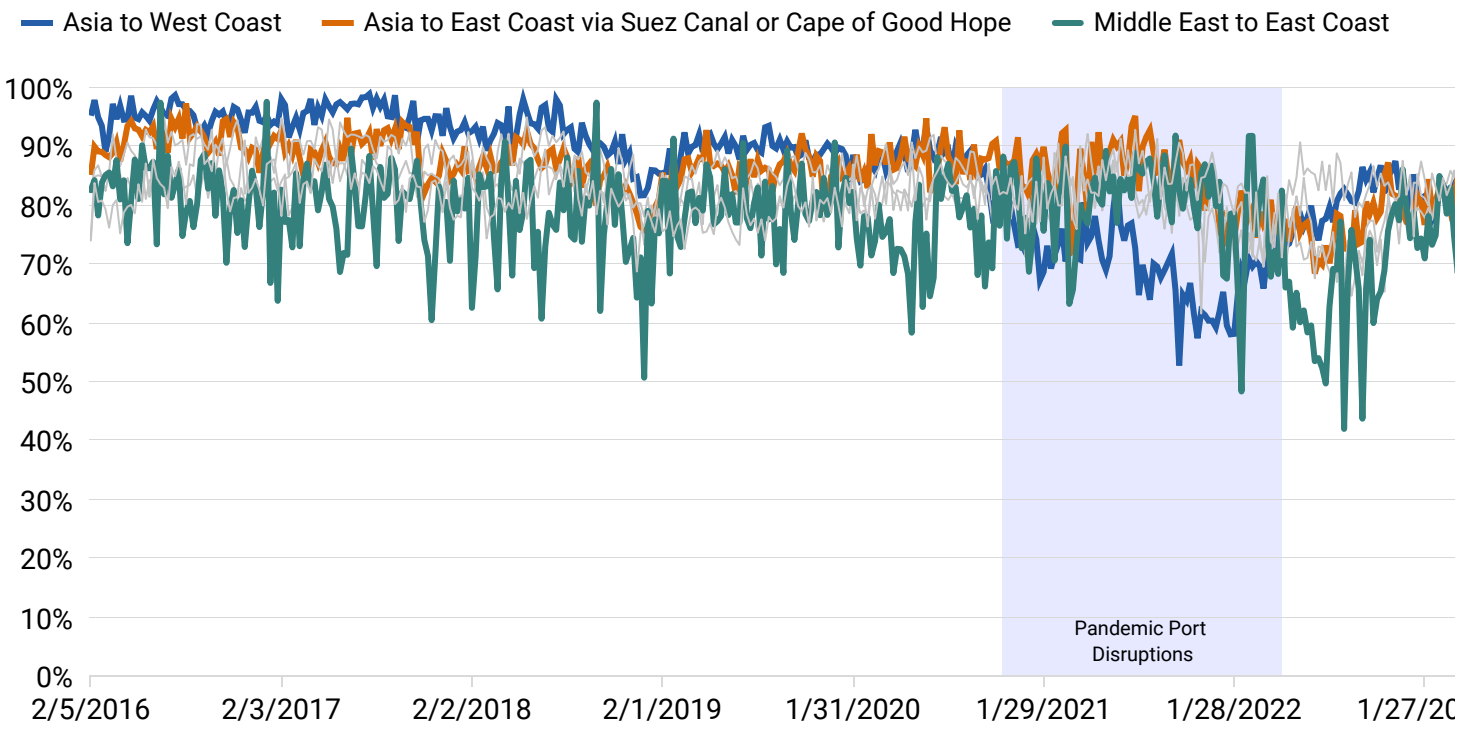
to 800 extra nautical miles for every week at sea, and ships from the Middle East, 1,000 to 1,250 extra nautical miles per week.

These additional miles used scarce vessel capacity without delivering additional cargo. In 2024, rerouting absorbed about 25% of potential capacity on the Asia-East Coast route via the Suez and about 36% on the route from the Middle East to the U.S. East Coast. As a result, shipping utilization fell sharply. On the Asia-East Coast route through the Suez, utilization fell by about 21 percentage points between March 2023 and March 2024, from percentages in the mid-80s to the low 60s. In other words, the share of potential shipping capacity converted into effective services fell by roughly one-quarter. By the end of our sample, utilization was about 61% on that Asia-East Coast route and 48% on the Middle East-East Coast route.

The causes of the pandemic disruption and Red Sea disruption were very different. In 2021, ships lost capacity primarily because of congestion and slower sailing. During the Red Sea crisis, ships lost capacity primarily because they had to travel farther. Measuring these losses in terms of effective shipping capacity allows us to compare them using the same metric.

Two Crises, One Measure of Shipping Utilization

Share of Potential Capacity turned into Effective Capacity



SOURCE: Bai, Fernández-Villaverde, Li, Marto and Zanetti (2026).
NOTES: Weekly data. Gray lines show other three major routes to the U.S. (from Asia via Panama, from Europe and from South America).

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Conclusion

Satellite vessel-tracking data provide a way to observe shipping disruptions as they happen and to measure how much of the fleet's potential capacity is actually available to transport goods.

The distinction between potential and effective capacity is important. During the 2021 port crisis, shipping companies responded to high U.S. import demand by deploying substantially more vessels toward the United States. Yet effective shipping services did not increase proportionately because congestion and slower sailing absorbed much of that additional capacity. During 2024-25, rerouting around the Red Sea reduced effective capacity even without large queues of ships outside U.S. ports.

These disruptions also do not necessarily remain confined to the routes where they begin. Shipping companies can respond by reallocating vessels across routes, while importers can change where they source their goods. Understanding these responses is important for determining how a disruption in one part of the shipping network affects trade, prices and economic activity elsewhere.

Notes

1. See Marcus Hand's Oct. 21, 2021, *Seatrade Maritime News* article, "Record 79 Containerships Queued at LA/Long Beach Ports."
2. See our working paper "Shipping to America," NBER Working Paper No. 35808, September 2026.

ABOUT THE AUTHOR



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Ricardo Marto is an economist at the St. Louis Fed, which he joined in August 2023. His research focuses on topics in macroeconomics, economic growth, industrial organization, and labor economics.

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